

LAYER 5 SPECIFICATION · GATED FUSION & FORENSIC PORTAL

Layer 5: Forensic Verification Portal & Gated Fusion Engine

Note (Oct 2026): superseded by [ARCHITECTURE.md](#), which documents the current gating rules (virtual capture device and failed sensor challenge now BLOCK; blind PRNU is advisory; files can never be labelled an authentic live stream).

Component: Layer 5 (Frontend & Integration Orchestrator)
Architecture: CEN/TS 18099 Gated Presentation Attack & Injection Detection
Target Latency: Sub-5 seconds end-to-end verification

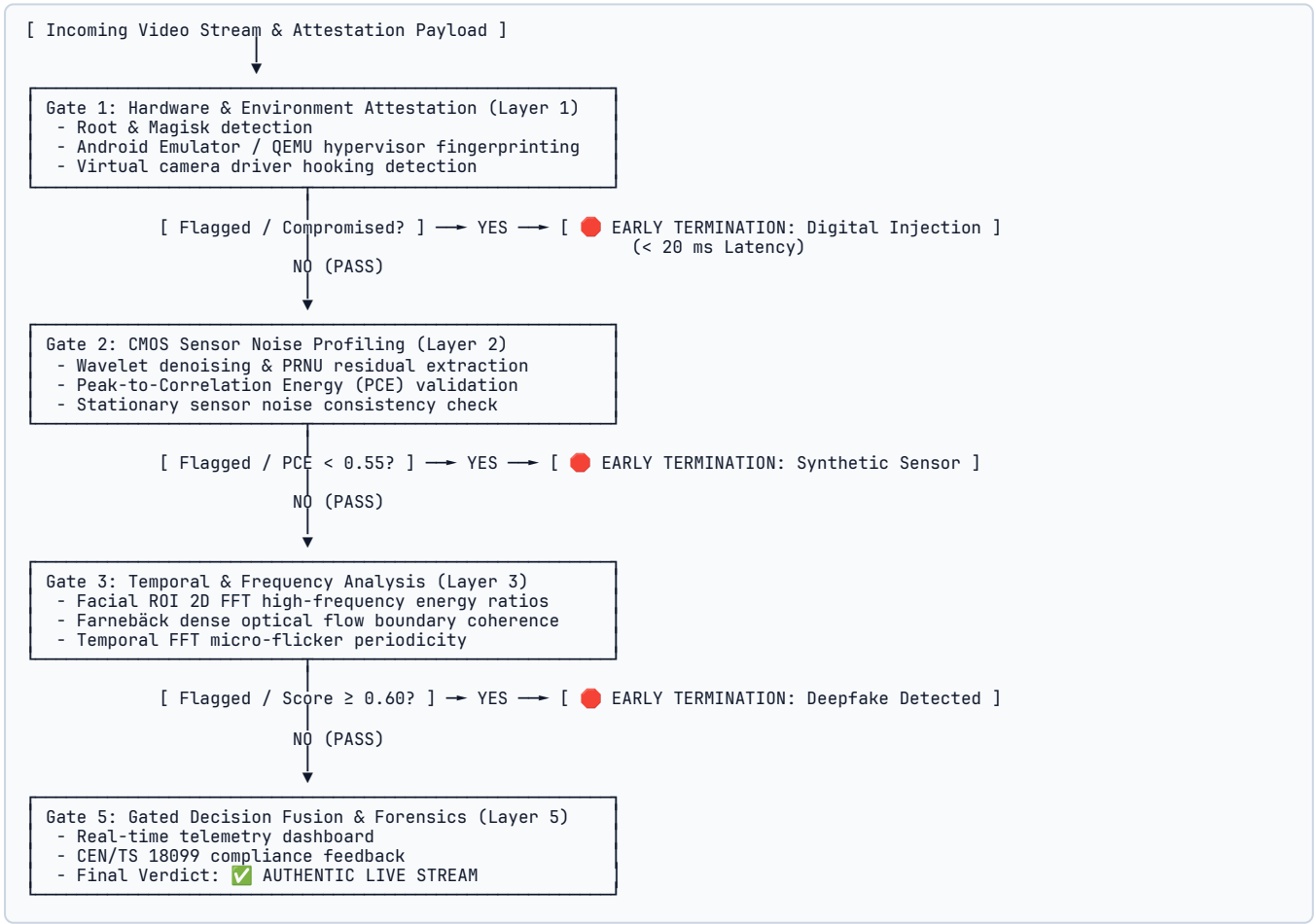
Overview

This module implements Layer 5 of the five-layer detection pipeline. It provides a real-time Streamlit forensic dashboard and an orchestration engine that unifies client-side attestation, CMOS sensor noise profiling, and temporal consistency checks into a sequential, gated execution flow.

Rather than running heavy spatial, frequency, and optical flow transformations simultaneously across entire video files, the pipeline enforces **early termination**:

- Failure at any intermediate gate halts execution immediately to preserve computational resources.
- Downstream mathematical transforms run only if preceding hardware and physical sensor criteria pass.

The 5-Layer Gated Architecture



Performance & Optimization Engineering

- Standard forensic video algorithms often require 1–3 minutes per video when processing every frame. To meet real-time operational requirements:
- Dynamic Burst Pre-Sampling: Implemented `create_fast_sample_clip` to extract a 45-frame burst (~1.5 seconds of video) for heavy mathematical analysis (PRNU wavelet decomposition and spatial/temporal FFTs). This reduces total verification latency to ~3 to 5 seconds.
- Zero-Compute Early Rejection: Compromised Android environments (Root/Emulator) are halted at Gate 1 in < 20 ms, eliminating unnecessary CPU/GPU load.
- Granular Forensic Telemetry: Surfaced raw metrics—including PCE Peak Energy, PRNU Anomaly Score, Attestation Latency, and Temporal Flicker—for full forensic explainability.

File Structure

```
├── app.py                # Streamlit forensic dashboard & telemetry display
├── pipeline.py           # Layer 5 orchestration engine & dynamic frame sampler
├── camera_sensor_noise_profiling.py # Layer 2 PRNU profiler (Arihant)
├── temporal_consistency_analysis.py # Layer 3 Temporal/FFT analyzer (Vibha)
├── generate_test_videos.py # Synthetic video generator for validation testing
└── README.md            # Pipeline documentation
```

Getting Started

Prerequisites

Ensure the required dependencies are installed:

```
pip install streamlit opencv-python numpy scipy
```

Running the Dashboard

Launch the interactive portal:

```
streamlit run app.py
```

- Open <http://localhost:8501> in your browser.
- Select the client device attestation state in the sidebar (to test hardware gating).
- Upload a video clip (.mp4, .mov, .avi) or test synthetic clips (fake_sim.mp4 / real_sim.mp4).
- Click Run Forensic Verification to inspect real-time gated execution and forensic metrics.